WEB DATAMINING AND SEMANTICS PROJECT

LEBANON AND ITS CULTURE

Eva BERRY and Charles AYOUB
TD DIA 1

Summary

I) Introduction *2*

II) Data Acquisition & IE *2*

II.I) Domain and seed URLs **2**

II.II) Crawler design and ethics **3**

II.III) Cleaning and NER **3**

II.IV) Ambiguity cases **4**

III) KB Construction & Alignment *4*

III.I) RDF Modelling choices **4**

III.II) Entity linkage **4**

III.III) Predicate alignment **5**

III.IV) Expansion strategy **5**

IV) Reasoning (SWRL) *7*

V) Knowledge Graph Embeddings *7*

V.I) Data cleaning and splits **8**

V.II) Two models **8**

V.II.I) TransE embedding model : explication and parameters 8

V.II.II) TransE model results 9

V.II.III) ComplEx embedding model: explication and parameters 10

V.II.IV) ComplEx model results 10

V.III) Comparison of the models **11**

V.IV) Size-sensitivity (20k / 50k / full) **12**

V.V) Nearest neighbours **13**

V.VI) T-SNE **14**

VI) Critical reflexion *16*

VI.I) Impact of predicate alignment quality 16

VI.II) Impact of noisy expansion 16

VI.III) Effect of ontology modeling choices 16

VI.IV) Open-world assumption vs embedding assumptions 16

VII) RAG over RDF/SPARQL *17*

VIII) Conclusion *24*

Report web datamining

Link of the GitHub: https://github.com/evaberry142-dev/v1.0-final.git

I) Introduction

Nowadays, the world keeps evolving and become bigger day by day. It's important to put interest in other countries' habits and try to discover as much as possible new cultures. In fact, it's not only important for our personal culture but also for our professional life, as we can be lead to collaborate with international people.
As part of the web data mining project, we wanted to take this opportunity to share our attachment towards our originary and lovely country: Lebanon, the Swiss of the middle east.

In this study, we will have the opportunity to discover the history of Lebanon, its culture, the places and most famous cities to visit as well as the friendship between France and Lebanon.
N
So, in order to conduct this study, we have chosen few websites that displays those informations and we are going to use web scrapping to explore them, as well as the SpaCy and NER tools in order to filter those pages. Finally, we will use the RDF graph in order to feed our chatbot based on an ollama model.

II) Data Acquisition & IE

II.I) Domain and seed URLs

In order to explore the graph of our subject, we need to extract the data. For that, we decided to choose sources and more specifically websites about Lebanon and its culture.
First of all, we decided to take websites that talks about "Beirut", the capital city of Lebanon. In order to extract relevant and detailed information about it, we used a Wikipedia page:

- Beirut Wikipedia page: https://en.wikipedia.org/wiki/Beyrouth

Still about the diversity of cultures in Lebanon, we extracted the Wikipedia pages of few famous cities in Lebanon, such as:

- Tripoli, the second bigger city in the country: https://en.wikipedia.org/wiki/Tripoli,_Lebanon

- Jezzine : https://en.wikipedia.org/wiki/Jezzine

In addition, we decided to extract informations about the rich history of Lebanon:

- History of Lebanon, Wikipedia page: https://en.wikipedia.org/wiki/Histoire_du_Liban

Finally, after seeing the diverse cities and their cultures, we needed to share the culture of the patrimonial activities such as the Mim-Museum, a jem and jewelry museum located in Beirut:

- Mim-Museum article extracted from “artshelp” website: "https://www.artshelp.com/mim-museum/"

Now that you know more about Lebanon and its beautiful culture, as a person living in France, you must know about the friendship between France and Lebanon. Indeed, the relationship between those two countries is very deep and last from a long time now.

- Relationship between Lebanon and France: https://en.wikipedia.org/wiki/France%E2%80%93Lebanon_relations

II.II) Crawler design and ethics

Now that we have chosen our sources, we need to extract the content of the pages. In order to do that we used the “Trafilatura” library in order to fetch and script HTML. By using this library, we will be able to take only the main content of a page, without the ads and images.

However, as we want to have as much as possible and relevant data, we fixed a minimum of 500 words on the page. Moreover, still talking about the relevance of the data, we needed to clean the data in order to be able to exploit it in the next parts. This is why for each URL, we avoid the duplicates and then we download the HTML page by making an HTPP request. Then, we also extracted the title using a "get" request from the "data" json that contains the website content. However, if this action fails, we used the "BeautifulSoup" library in order to transform the HTML into a readable tree so that we can then extract the title. Finally, we saved the title, content and URL of each website in the json file “crawler_output”.

II.III) Cleaning and NER

Once we have cleaned and prepared the data, we needed to start the filtering of the data. First, we have loaded an NLP pipeline in order to filter the text in English language using the spacy model.
In addition, in order to form an RDF graph in the next parts, we needed first to extract the entities of our subject. This is what we have done by using our jsonl file "crawler_output" that contain the data and we transformed each line into a dictionnary. Then, we use Spacy (NLP model) in order to analyze the extracted text and for each entity we assign to it a category that match between: ["PERSON", "ORG", "GPE", "LOC", "DATE"].

Finally, we cleaned the texts, and we deleted the spaces and the small words such as "the" "so" ect. Indeed, those words are not going to be relevant for our study. Moreover, we will keep only the entities that have an uppercase at the beginning of the entity because usually the entities take that form. In addition, we will keep only one

word if we have duplicates. Once we finished all those steps, we saved our entities in a file name “extracted_knowledge.csv”.

Furthermore, once we had our entities, we needed to add information about it and form the subject-verb-predicate form. In order to accomplish that, we used again the "crawler_output" jsonl file and we extracted the verbs. Then, based on the verb we tried to catch the subject and the object. Finally, everything has been stored in a csv file named "relations.csv".

II.IV) Ambiguity cases

However, when we extracted our entities and we tried to classify them into the different categories: person, organization, GPE, location and data, we noticed some ambiguities. Indeed, informations such as food were classified as persons because there are no food columns. In order to handle the misunderstood of the food recognition, we did not take pages that deals only about Lebanese food. We only kept the Wikipedia pages about the cities, universities, museums and the relationship between Lebanon and France.

Moreover, we also noticed that some words that are not in French were not selected or not classified correctly. Another important point about the classification was that on the one hand, “Beyrouth ville” has been classified as a Geo-Political entity. On the other hand, “Beirut” has been classified as a “LOC”. In fact, both classifications are correct because, besides being a city and therefore a location, “Beyrouth” is a city with a political lead (capital of Lebanon) and has a renown administrative entity, this is why we can classify it as an GPE.

III) KB Construction & Alignment

III.I) RDF Modelling choices

Now that we have our data and relations between the entities, we can finally create our RDF Graph. In order to do that, we translated the relations by creating URIs for each entity and its respective subject, predicate and object. Finally, all the URIs has been stocked in the "knowledge_base.nt" file.

III.II) Entity linkage

Once we had created our knowledge graph, we had our disposal a large database about Lebanon history and culture. The next step was to relate those entities to their equivalent on WIKIDATA, a website that allows us to associate our entities with unique identifiers (QIDs). In order to do such enrichment and equivalence of identifiers, we used the API of WIKIDATA.

However, in order to improve matching quality, we normalized the entities by putting everything in lowercases, deleting special characters and additional unnecessary spaces. In addition, we also ensured to generate valid URIs for our entities.

Furthermore, once we had done that, we started the API calls on Wikidata and we filtered our request so that we can retrieve for each entity five candidates. Indeed, sometimes, for some entities, we had few possibilities and therefore we chose to keep only five of them each time. However, as we chose to keep five candidates, we needed to find a solution to differentiate them and choose the best one. This is why we used the ”SequenceMatcher” in order to compute a similarity score between the entity that we have in our database and the candidates found in wikidata.
In fact, the "SequenceMatcher" library compares words by words the two under-common sequences. Each time, we return a score of similarity between "0" and "1". The computation of this ratio takes the following form:

Similarity = (2*M)/T

with M = number of common characters
and T = total length of the two characters sequences

Moreover, for each entity that have a equivalent on wikidata we tried to recuperate its label and description and in the case where we have a description, we will added a 0.05 on its confidence score because it's a trustfull match. Indeed, for each entiry, we usually have few candidates on WIKIDATA, so that's why we applied the score process on those candidates each time and we selected the one that have the best confidence score.
Finally, we stored this mapping in a file named “entity_linking_mapping.csv”. Most precisely, we put in this file for each entity, its name, wikidata URI, confidence score and finally its label and description if there is one.

At the end of the linkage, we made sure to integrate the aligned entities to the knowledge graph using “owl:sameAs” links between the local entity URI and the corresponding Wikidata entity.

III.III) Predicate alignment

Once we had aligned the entities, we needed to do the same for the properties. In order to do that, we have extracted the properties of the RDF graph and then for those who have a subject and an object that are aligned with WIKIDATA, we extracted them using a SPARQL query the candidate properties from WIKIDATA. The final alignment is then obtained through a manual validation step.

Each time, we extracted the eligible triples. In other words, we extract the triplets that have aligned subject and object. If the subject and the object were not aligned in the entities file (with an equivalent in WIKIDATA), we did not extract them. Following that, we extracted the eligibles we made queries to WIIKIDATA to see their equivalent on it in order to extract them.

III.IV) Expansion strategy

In the previous part of the project, we searched for the eligible predicates (the one that their aligned subject and object are available in the entities found in wikidata) on WIKIDATA. Therefore, we succeded to extract some triplets.
However, in order to have a proper and rich database, we needed to enrich it. This is why we used the KB expansion via SPARQL in order to enriche our database using data contained in WIKIDATA. For that, we used the One Hop expansion and the Two Hop expansion.

The principle is that the One Hop expansion is a principle based on a query that will allows us to find and extract the neighbors of each of ours entities available on wikidata. In that way we will be able to have more data about our subject.

Concerning the Two Hop expansion, we will use the relations extracted in the One Hop expansion in order to generate their respective relations. Once again, we will have a more enriched graph as we will have the relations of the relations in addition of the original relations obtained from the articles and their wikidata equivalent.

Another important point of the hop expansion, is the parameters of the SPARQL queries. We fixed some limits to the one hop and two hop expansions. Indeed, the One Hop can have thousands and thousands of relations, so we will limitate it to 200 as otherwise we have the risk to explose our graph. It's the same for the Two Hop that can generate even more relations as the neighbor of the neighbor can really become a very large graph. This is why we fixed a limit of 40. Moreover, we also fixed a limit of 15 as it will reduce the code cost and avoid the reduce of the efficiency.

Once we have done with the enrichment using the hop expansions, we added the new triplets in our RDF Graph. After enrichment, we succeeded to have the following numbers:

```
Final triples: 63370
Entities: 10537
Relations: 2046
```

With such results, we are in the right range of values which is very good for our study. However, we have 2046 relations which exceed a lot the range of 50-200. In order to reduce the number of relations, we will extract the most frequent relations and keep the 150 most fequent ones.

After filtering, here are the finals statistics:

```
Filtered triples: 58256
Entities: 7449
Relations: 150
```

Now, everything is mostly in the right range.

IV) Reasoning (SWRL)

In this part we will compare the rule-based method and the embedding base reasoning.

First, we made a test of the rule-based method on the file “family.owl”. The goal was to test the principle of the rule-based and the reasoner on a true ontology file. This “family.owl” file is composed of an ontology with normally classes et individuals. The goal was to apply SWRL a rule that state "a person who is older than 60 years old is an 'oldPerson'" ('oldPerson' is a class). In order to do that, we need charged the "family.owl" file and downloaded the the OWLReady2 in python. This allowed us to charge the ontology contained in the 'family.owl" file.

In order to apply our rule, we first checked the content of the classes, individuals, object properties and finally data properties. While we were checking those informations of the family.owl file, we remarked that there were no individuals. This is why, we added manually using python command some individuals.

Finally, once we had individuals in the family.owl file and we have checked that we have the class “Person” and the property of the age, we added the SWRL rule with the condition of the individuals older than 60 years old. Once this query was set up, we need to launch the reasoner so that we can apply the rule and therefore the reasoner will add the consequences of this rule to the ontology.

Furthermore, as mentioned just before, in the SWRL method, we define a rule and then the reasoner make deductions. Here, in our project, we can propose this rule:

- BornIn(?p, ?v) ∧ isLocatedIn(?v, "Lebanon") → BornIn(?p, "Lebanon")

In contrast, in the embedding based reasoning method, we did not fix a rule. We only gave a lot of examples to our TensE model. The AI make the process alone and it goes from a person to a city and then to a country. It's simply the equivalent into vectors, such as :

Vecteur(BornIn) + Vecteur(isLocatedIn) = Vecteur(BornInGlobal)

So, in reality, with the embedding method, we achieve to the same conclusion thanks to the geometry. Finally, we can say that the embedding can find informations without having them in the database even if he is a little blurred. The embedding appears to be a simplified version and less strict version of the SWRL method.

V) Knowledge Graph Embeddings

Now that we have our RDF graph that is ready, we need to apply embedding models in order to transform entities and relations into vectors in order to catch patterns in our graph.

V.I) Data cleaning and splits

Before applying any embedding model on our data, it's necessary to clean and prepare it.
First, we simplified the URIs so that we can avoid the long ones. In fact, we recuperated the last member of each URI. Moreover, we also checked and removed the duplicates in the triplets. Once we have done with those checks, we stocked all the finals triplets in a file named "cleaned_triples".

Furthermore, one we have done with the preparation, we need to split the data into training, testing and validation, so that we can be able to apply our embeddings models. This is what we have done in our project. We fixed the repartition of the data as follow: 80% for the train, 10% for the validation and 10% for the test. Moreover, we also make sure that there is no entity that appears only in the validation/test set. Those three datasets were finally stocked in the following files: "train.txt", "valid.txt" and "test.txt".

V.II) Two models

Once we have done with the embedding preprocessing, we can implement our embedding models. For that, we will choose two of the recommended models in the statement : TransE and ComplEx. Indeed, we can easily compare those two models. On the one hand, TransE is a transational model where the entities and relations are vectors. On the other hand, ComplEx is good as it handles asymetric relations.

However, in order to implement those two models, we need to have numerical values. For that, we will use the PyKEEN framework, as mentionned in the statement. This tool will allows us to convert our data and index it. In that case, each entity will have its own numerical ID. Moreover, PyKEEN will stock them in a dictionnary (mapping phase).

V.II.I) TransE embedding model : explication and parameters

As mentioned above, for the first embedding mode, we chose the TransE model.

In order to implement our models, we needed to fix some parameters. However, as the goal was to compare the two embedding models, we decided to fix the same parameters for both of the models.

Concerning the parameters choice, we chose a dimension of "100" because it allows us to avoid over-fitting. Indeed, a high value of dimension can lead to over-fitting. Moreover, such a dimension, can allows us to capture a big variety of semantial caracteristics without being very complex. Moroever, for the learning rate, we chose the value "0.001" which is equivalent to the "Adam" value and its considered as one of the most efficient prameter, it's also the one used by default by PyKEEN. In addition, we fixed our batch size to 128, which means that each time we will take 128 triples and this will allows us to have a stable gradient. Finally, we will fix the number of epochs at 50 so that we can have a good balance between efficiency and time-consuming for our model training. The negative samples is fixed to "10" which means

that we present 10 fake examples for only 1 true example. This will avoid the trivial answers of our model and make it more robust.

V.II.II) TransE model results

Now, let's analyze the results of TransE model.

Results for both head and tail:

```
TransE - MRR: 0.2047077864408493
TransE - Hits@1: 0.11854838709677419
TransE - Hits@3: 0.22903225806451613
TransE - Hits@10: 0.38870967741935486
```

Figure 1: Picture of the results of the TransE model for the head and tail

According to the results above, we see that the model has an average performance with an MRR of 20% which is a good result. This means that in average the correct entity appears in the fifth rank.
Concerning the Hits@K scores, we see that the model tends to have difficulties in finding correct answers in the top 1. However, in the top 3 and top 10, TransE model succeeded to have a better number of correct entities with a score of 22% for the Hits@3 and 38% for the Hits@10.

Now, let's analyse the results for the head:

```
Head - MRR: 0.20255686342716217
Head - Hits@1: 0.12741935483870967
Head - Hits@3: 0.22419354838709676
Head - Hits@10: 0.3629032258064516
```

Figure 2: Picture of the results of the TransE model for the head only

According to the results of the head prediction, we see that the MRR is the same as for the general result (both head and tail). So, the model is consistent between his different tasks. Moreover, we see that the Hits@1 score is better for the head with a score of 12% comparing to 11% for the general. However, we have a little decrease for the Hits@10 which suggest that the model have less correct entities in the top 10 for the head.

Now let's see the results for the tail:

```
Tail - MRR: 0.20685870945453644
Tail - Hits@1: 0.10967741935483871
Tail - Hits@3: 0.23387096774193547
Tail - Hits@10: 0.41451612903225804
```

Figure 3: Picture of the results of the TransE model for the tail only

Here, for the tail, the MRR is a little bit above the head with an increase of 0.6. The model tends to classify better the right tails. However, the Hits@1 score descreased and is about 10% which means that the model lack of precision on the correct candidates in the top 1 for the tail.
Finally, the model achieved good results for the top 10 as he succeeded to obtain a score of 41% of correct entities.

V.II.III) ComplEx embedding model: explication and parameters

Concerning the parameters, as mentioned before, we decided to keep the same ones as for TransE model, so that we can fairly compare them.

V.II.IV) ComplEx model results

First, let's analyse the global results for both the head and tail.

```
ComplEx - MRR: 0.010800539515912533
ComplEx - Hits@1: 0.004032258064516129
ComplEx - Hits@3: 0.013709677419354839
ComplEx - Hits@10: 0.016935483870967744
```

Figure 4: Picture of the results of the ComplEx model for the head and tail

According to the results above, we clearly see that the ComplEx model have a lot of difficulties in catching the correct entities. The MRR is extremely low with a score of 0.01. This suggest that the average rank of the correct entity is about 100 which is not good at all. This means that the model does not learn correctly.
In addition, the Hits@K scores (top 1, top 3 and top 10) also confirms this bad classification. Indeed, mostly all the scores are close to zero.

```
Head - MRR: 0.010624585673213005
Head - Hits@1: 0.0032258064516129032
Head - Hits@3: 0.014516129032258065
Head - Hits@10: 0.016129032258064516
```

Figure 5: Picture of the results of the ComplEx model for the head

According to the results of the head, we constate approximately the same results as for the general version (head and tail). This reinforce the idea that the ComplEx model is not efficient.

```
Tail - MRR: 0.01097649335861206
Tail - Hits@1: 0.004838709677419355
Tail - Hits@3: 0.012903225806451613
Tail - Hits@10: 0.017741935483870968
```

Figure 6: Picture of the results of the ComplEx model for the tail

According to the results of the tail, we constate approximately the same results as for the general version (head and tail) as well as the head version. This reinforce more in more the idea that the ComplEx model is not efficient. We even have scores equal to zero!

V.III) Comparison of the models

Now that we have analysed each model separately, we can now compare them.

Here is the resume for the results in both head and tail for the two models:

	model	MRR	Hits@1	Hits@3	Hits@10
0	TransE	0.204708	0.118548	0.229032	0.388710
1	ComplEx	0.010801	0.004032	0.013710	0.016935

Figure 6: Picture of the results of the TransE and ComplEx model for the head and tail

	Model	MRR	Hits@1	Hits@3	Hits@10
0	TransE	0.202557	0.127419	0.224194	0.362903
1	ComplEx	0.010625	0.003226	0.014516	0.016129

Figure 7: Picture of the results of the TransE and ComplEx model for the head

	Model	MRR	Hits@1	Hits@3	Hits@10
0	TransE	0.206859	0.109677	0.233871	0.414516
1	ComplEx	0.010976	0.004839	0.012903	0.017742

Figure 8: Picture of the results of the TransE and ComplEx model for the tail

Interpretation of the results:

According to the results above, we clearly see that the TransE model is better than the ComplEx model. This result is valid for all the metrics : head and tail together, head alone and tail alone. Indeed, in all the metris, the MRR is about 0.20 for the TransE comparing to only 0.01 to ComplEx. This suggest that using the TransE model, the correct entity is usually the 5th position in the results.

Moreover, concerning the Hits@K scores, which correspond to the pourcentage of time the right answer is in the top K, the TransE model outperform ComplEx model too. Indeed, for the TransE, the right entitiy is in top 1 in 10% (11% for both, 12% for head and 10% for tail) of the time whereas in the ComplEx model, it's present in only 0.004 % of the time.

In addition, we also see the difference between the two models while we analyze the difference between the head and tail. Indeed, on the one hand, for the TransE model we have a good balance between the head and tail for the MRR with 0.2026 and 0.2069 respectively. However, it still have a very small gap for the Hits@10 that is higher in the tail with a pourcentage of 41% comparing to 36% in the head. this show us that the model predict better the objects than the subjects.
On the other hand, for the ComplEx model, the head and tail are both very low (around 0.01) and therefore we does not have a useful learning.

Overall, we can say that the TransE model outperform the ComplEx model.This can be due to the fact that we kept the most frequent relations and therefore we simplified our graph. Moreover, TransE model is very good at catching simple structures and allows us to have good results here. On the contrast, ComplEx model has a lot of difficulties in performing and this can be explained by the size of our dataset or noise. If we cant to improve the result of the ComplEx mode, we could try to adapt the parameters(tuning).

V.IV) Size-sensitivity (20k / 50k / full)

Now that we have see that TransE model is better than ComplEx model, this is why we decided to divise our KB database into three dfferent subsamples : 20K triplets, 50k triplets and the full dataset. For each sample, we applied our TransE model that we have created just before.

Here are the results for the three models with each time a different size of samples:

```
Results of sensitivity analysis:
KB_Size      MRR   Hits@1   Hits@3  Hits@10
    20k 0.135762 0.065323 0.145968 0.281452
    50k 0.125925 0.070968 0.125806 0.241129
   full 0.113670 0.054032 0.119355 0.237097
```

Figure 9: Picture of the results of the TransE model for three different samples size

According to the results above, the 20k sample dataset is better. Indeed, it achieved the better scores in all the metrics with 13% for the MRR, 6% for the Hits@1, 14% for the Hits@3 and finally 28% for the Hits@10, which is higher than the full dataset even if it still very close to the full dataset which indicate that the full dataset is also one of the best.

However, we notice that the dataset with 20k samples is better than the one with 50k. It's sound quite unconsistent with the fact that the full dataset is better and therefore the more we have data, the more we have good result. I other word this disharmony, suggest that increasing the dataset size does not always lead to better performance. This can be explained by the noise of the data. Indeed, maybe the dataset with 50k samples introduces triplets that are less reliable and the relations less pertinent.

Overall, those results show us that the quality and quantity of the data is very important for our knowledge graph embedding performance. So, one need to be careful while choosing the data filtering and sampling strategies.

V.V) Nearest neighbours

Now that we have finished the models, we tried to recuperate the neighbors of each entity in the embedding space. In that way, we will be able to see if the TransE model (our better model) succedeed to regroup in the vector space, the entities that are really similar.

Concerning its process, we first extracted matrix that contains the vectors of all the entities in 100 dimensions as well as the triples used in the training.

Then, we iterated on each relation, and we identify for each relation its associate subject and we filter the very small categories (less than 3 entities) so that we can have a sgnificant statistical mean.
Following that, we calculated the centroids which are the points of gravity where the entities should convgerge to it. In addition, we computed the coherence using the Similar cosine. In fact, for each entity of a group, we calculated the angle between the vector and its centroid. If the score is equal to 1, then it means that we have a perfect

match of the entity with its category. Finally, we measured the mean of those scores in order to have the coherence rate of the relation.

Here are the obtained results:

Analysis of the semantic coherence

Category (Relation)	Number of Entities	Coherence Rate (Cosine)
P5572	4	0.6733
P1476	3	0.6431
P684	8	0.5592
P150	7	0.5000
P373	6	0.4689
P190	6	0.4437
P39	14	0.4309
P166	9	0.4163
P269	14	0.4108
P2894	31	0.4081
P69	18	0.4067
P101	10	0.4006
P734	13	0.3974
P460	11	0.3958
P1412	19	0.3908
P20	20	0.3902
P26	17	0.3896
P463	10	0.3876
P47	11	0.3865
P40	15	0.3863
P19	28	0.3699
P112	15	0.3691
P27	42	0.3637
...		
http://schema.org/version	847	0.1228
http://www.w3.org/2000/01/rdf-schema#label	655	0.0788

Global Coherence rate of the model: 0.3437

Figure 10: Picture of the results of the nearest neighbors

According to the results above, the TransE model succeded in creating solid geometrical clusters with a score of 67% for the relation "P5572" and 64% for the relation "P1476". This good result for those two entities can be explained by the fact that we have only 4 and 3 entities in those categories which facilitate their regroupment.

In addition, if we look at the evolution, we see that the more the number of entities increase, the more the coherence rate decrease. This is due tot he fact that the more we have diversity in the data, the more TransE will have difficulties to keep a big number of entities close to one centroid.

Finally, we see at the end, that the general relations such as the labels and versions have a very low coherent rate (12% and 7%) which is consitent with the fact that these relations do not define a precise semantic category.

V.VI) T-SNE

Once we have seen the number of entities in each cluster (relation) and its coherence rate, we did a visualization that plot those clusteers. In order to do that, we used the tool t-SNE. This tool allows us to reduce the original dimension (100) into two dimension and therefore we will be able to plot the clusters.

(Note : I tried to apply the TSNE on all the entities but it crashed all my kernels and I had to re run all my project many times. There, after few attempts I reduced the t-SNE to 1000 entities because it was the only supported value by my computer)

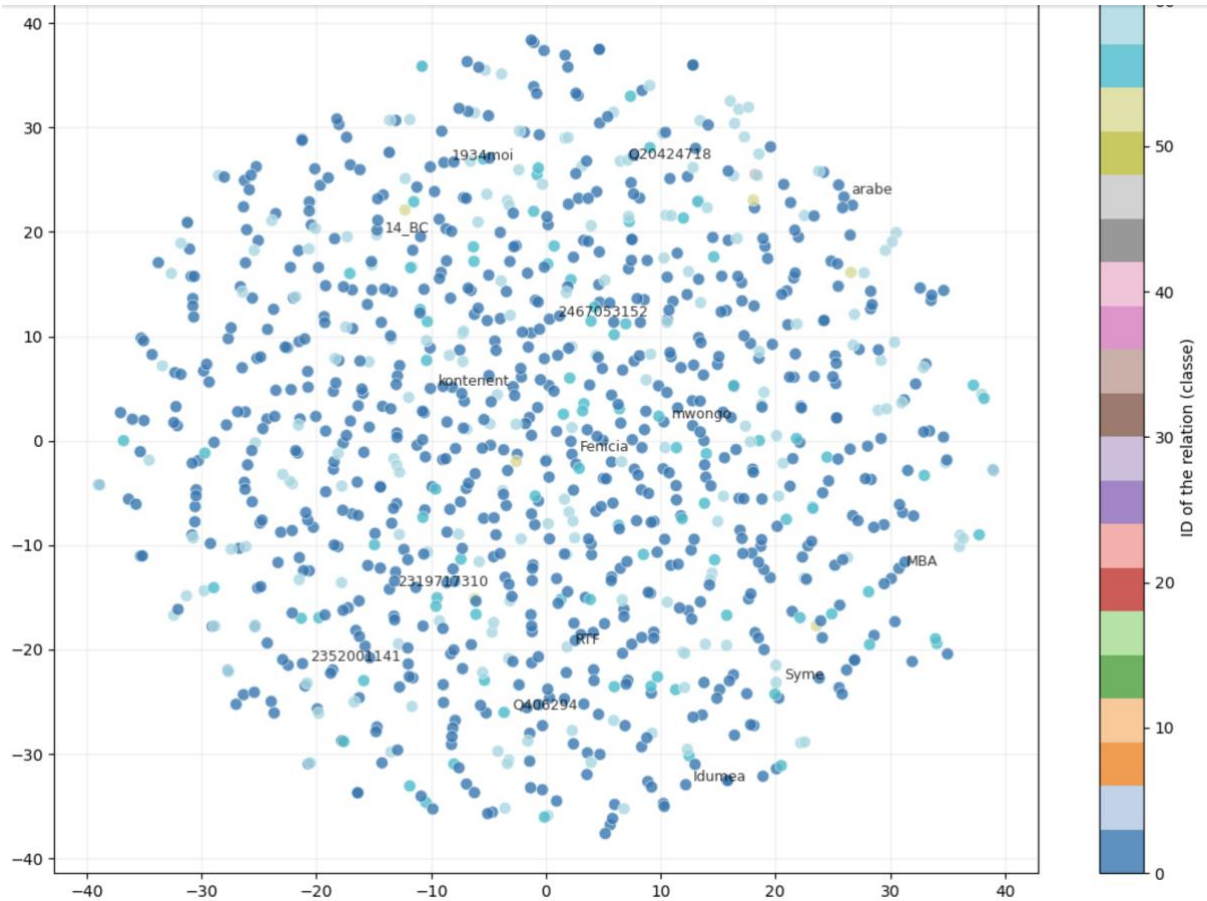

Figure 11: Picture of the results of the nearest neighbors

According to the graph obtained above, we see that the relation ID are spread in micro-groups of identical colors that are very close together. For example, dark blue dots or the turquoise dots tend to "travel" together in small clusters. However, the regroupment is not clearly defined and perfect. This suggest that the entities in our dataset are multimodal, and some entities can belong to few concepts.

Moreover, the embedding captured well the semantic structure. Indeed, as we can see on the graph, we have few places with different colors and we clearly see the difference in the location of the different entities. For example, we have the region "Idumea" and the language "arabe" that are located in precise locations in the graph. Thus, show that the model succeeded in creating logic (sense of the relations) with the geometry (the position in the space).

Moreover, we also see the micro-clusters on the graph. Indeed, here, the entities that belongs to the same class are not spread in the graph and they form micro-clusters. We can see that thanks to the mini regroupment of colors each time. This allows us to say that the model understood that those entities have properties and informations in

common. However, we can note that we still have some entities that are overlapping each other (which is normal as one entity can have few roles).

Finally, we also see a specific hierarchical and contextual context. Indeed, in the middle we have the most connected entities and in the extremities of the graph we have the most rare concept.

Therefore, we can say that our model succeeded in extracting the sense of our web pages.

VI) Critical reflexion

VI.I) Impact of predicate alignment quality

If we have some unaligned relations, we will create noise and therefore TransE model can create fake relations. This is probably what has been done on the entities that are at the boarders of the visualization. Moreover, a unaligned predicated create a bad mean rank for everyone.

VI.II) Impact of noisy expansion

In our project, we have probably added noise by enriching information in our base. Indeed, maybe it enriched our entity with similar words but maybe those words did not have the same meaning. This can create illogic schema in our graph and therefore this can have consequences on our vector embedding.

VI.III) Effect of ontology modeling choices

The ontology choice can have impacts on our study. Indeed, if we choose classes in a large way or in a very precise way, we can have embeddings that are located in the center. This is what se have seen in the graph of embedding visualization. Therefore, if the ontology is too much simple, it will make the entities converges in the center.

VI.IV) Open-world assumption vs embedding assumptions

In our project, TensE model need to extrac t the triplets and for some triplets it's going to generate fake examples by creating triplets that are not in our database. However, aside of our dataset there are triplets that are correct, it's just that they are not in our database. Therefore the model is going to those vectors (subject and object) move away from each other even if they are words that match together. For example, if we have, (Beirut, capital of, Lebanon) if does not exist in our database and the model consider it as a negative example, it's going to move away Beirut from Lebanon whereas those two words are in reality close !

Finally, the previous previous part of the project had an important impact on our project and especially the scraping and parsing phase and wikidata alignement phase. Indeed, if we missed capital letter, our entities can be duplicated and therefore our

model can create two different persons for the same entity. Moroever, if the triplets are not clean, we will not have a net t-SNE with structures and differents groups.

VII) RAG over RDF/SPARQL

Our system is based on the load of our RDF knowledge graph with N-Triples format. A schema summary is made by extracting the predicates of the graph and therefore it builds automatically a schema summary.

For this RAG implementation, we chose the LLM “Ollama” with the model “gemma:2b”. The principle is that we send a prompt to this model. Indeed, this prompt will allow us to guide the LLM. It contains three important elements: our schema summary, some examples related to Beirut, and finally rules. Those examples will allow us to show our model how to deal with the SPARQL query and it will give him an example. Concerning the rules that we have fixed in the prompt, we forbid the LLM to use prefixes such as wd: or wdt: and we imposed him to rely on the full URIs and generic variables such as ?p, ?s and ?o.

Here is a summary of this prompt:

“
You are a SPARQL expert. Use this schema summary:

=> schema summary

Examples:

[example questions and queries mentioned before]

Rules (mentioned before):
- Do not use 'wdt:' or 'wd:' prefixes.
- Use full URIs or use variables ?s ?p ?o.
- Output only the SPARQL code.

«

Finally, we make sure that only the SPARQL code is kept by applying a regular expression to extract the SELECT query.

In addition to the prompt, we implemented a self-repair mechanism which is basically the handle of the errors in SPARQL queries. We made a rule: the model make a first attempt to access the RDF graph and find the answer and if he fails, he goes back to the prompt so that he can try to generate the correct answer. This repairing system is repeated for three attempts each time he fails. Moreover, our code also allows the LLM to replace the invalid wikidata prefixes by replacing prefixes such as “wdt:*” and “wd:*” by variables.

Now let’s see some tests and results with our LLM and RDF Graph.

Each time, we plot the baseline, which means the answer of the LLM without the RAG and also the answer of the LLM using the RAG (search in our RDF graph the answer).

First test: “tell me about Beirut”

```
Loaded 58256 triples.

Your question (or 'exit'): tell me about beirut

--- BASELINE (LLM only, no KG) ---
Sure, here's a summary of Beirut:

Beirut is the capital and largest city of Lebanon, a country in the Middle East. It is known for its rich history, vibrant culture, and stunning archit
ecture.

**History:**

* Beirut has a rich history dating back to the 1st millennium BC.
* It was a major trading hub for over 2,000 years, and its port played a significant role in the development of the Mediterranean Sea trade route.
* In the 19th century, Beirut was the center of the Arab world and was a hub for political, cultural, and economic activity.
* After the Lebanese Civil War, Beirut became the capital of the Republic of Lebanon.

**Culture:**

* Beirut is a vibrant city with a rich cultural heritage.
* The city is home to a diverse population of people from all over the world.
* Beirut is known for its traditional music, dance, and cuisine.
* The city is also a center for art and culture, with a number of museums and galleries.

**Architecture:**

* Beirut is a city of stunning architecture.
* The city is home to some of the most famous buildings in the Middle East, including the Beirut Palace, the Umayyad Palace, and the Notre Dame Cathedr
al.
* Beirut is also a center for modern architecture, with a number of high-rise buildings and skyscrapers.

**Other:**

* Beirut is a major port city, with a busy commercial port and a large fishing industry.
* The city is also a major cultural and political center in the Middle East.
* Beirut is a popular tourist destination, with a wide range of attractions to choose from.
```

```
--- RAG (With your Knowledge Graph) ---
Processing SPARQL...

[Attempt 1] Executing:
?s a <http://example.org/relation/have>.
?s a <http://example.org/relation/include>.
?s <http://schema.org/dateModified> "2023-03-08T12:00:00Z".
?s <http://schema.org/description> "Beirut is the capital city of Lebanon and is located on the Mediterranean Sea in the Middle East.".
?s <http://schema.org/version> "1.0".
?s <http://wikiba.se/ontology#sitelinks> <http://example.org/beirut>.
?s <http://wikiba.se/ontology#statements> <http://example.org/statements>.
Error: Expected {SelectQuery | ConstructQuery | DescribeQuery | AskQuery}, found '?'  (at char 0), (line:1, col:1)

[Attempt 2] Executing:
SELECT ?s ?p ?o WHERE { ?s ?p ?o . FILTER(CONTAINS(LCASE(STR(?s)), "beirut") || CONTAINS(LCASE(STR(?o)), "beirut")) }

Final Answer (Grounded):
**Here's the information about Beirut from the RDF triples:**

* **Entity:** Beirut
* **Property:** label
* **Value:** 貝魯特
* **URL:** <https://dbpedia.org/resource/Beirut>

* **Entity:** West Beirut
* **Property:** sitelinks
* **Value:** 10
* **URL:** <https://dbpedia.org/resource/West_Beirut>

* **Entity:** the American University of Beirut 18
* **Property:** type
* **Value:** <http://www.w3.org/2000/01/rdf-schema#Resource>
* **URL:** <https://dbpedia.org/resource/American_University_of_Beirut_18>

* **Entity:** Q7294643
* **Property:** description
* **Value:** sector al Beirutului
* **URL:** <https://dbpedia.org/resource/Q7294643>

* **Entity:** Q722925
* **Property:** description
* **Value:** stadsdel i Beirut, Libanon
* **URL:** <https://dbpedia.org/resource/Q722925>
```

```
* **URL:** <https://dbpedia.org/resource/Q722925>

* **Entity:** Beirut River
* **Property:** P214
* **Value:** 315167842
* **URL:** <https://dbpedia.org/resource/Beirut_River>

* **Entity:** Beirut I
* **Property:** dateModified
* **Value:** 2026-02-27T11:11:48Z
* **URL:** <https://dbpedia.org/resource/Beirut_I>

* **Entity:** Beirut River
* **Property:** description
* **Value:** נהר
* **URL:** <https://dbpedia.org/resource/Beirut_River>

* **Entity:** Q1162983
* **Property:** label
* **Value:** Vilayet de Beirut
* **URL:** <https://dbpedia.org/resource/Q1162983>

* **Entity:** Beirut River
* **Property:** label
* **Value:** Afon Beirut
* **URL:** <https://dbpedia.org/resource/Beirut_River>

* **Entity:** Beirut
* **Property:** altLabel
* **Value:** பய்ரூட்
* **URL:** <https://dbpedia.org/resource/Beirut>

* **Entity:** Al Riyadi Beirut
* **Property:** altLabel
* **Value:** Ριάντι Κλαμπ Μπεϊρούτ
* **URL:** <https://dbpedia.org/resource/Al_Riyadi_Beirut>

* **Entity:** Q5646357
* **Property:** description
* **Value:** Stadtteil von Beirut (Sektor 34)
* **URL:** <https://dbpedia.org/resource/Q5646357>

* **Entity:** West Beirut
* **Property:** description
```

As we can see above, the model succeeded to give us an answer. Indeed, he gave us a link to the description of Beirut, West Beirut, the American University of Beirut (AUB) as well as places like sector al Beiruti and Stadsdel Beirut and stadium such as the “Al riyadi”.

Second Test: Tell me about university

```
Your question (or 'exit'): tell me about university

--- BASELINE (LLM only, no KG) ---
Sure, here's a summary of what a university is:

**A university is an institution of higher learning:**

* It offers undergraduate and graduate programs in various fields of study.
* It provides opportunities for research, internships, and other academic experiences.
* It attracts students from diverse backgrounds and prepares them for successful careers.
* It is typically funded by the government or through tuition fees.

**Key characteristics of a university:**

* It has a faculty of qualified professors and researchers.
* It offers a wide range of courses and research opportunities.
* It has libraries, laboratories, and other facilities for students to use.
* It has student organizations and activities that cater to different interests.
* It has a campus with buildings, residence halls, and other facilities.

**Benefits of attending university:**

* Higher education can lead to better career prospects and higher earning potential.
* It can also help you develop critical thinking, problem-solving, and communication skills.
* It can expose you to new ideas and perspectives.
* It can help you network with other students and professionals.

**Additional facts about universities:**

* They are typically located in major cities or metropolitan areas.
* They offer scholarships and financial aid to help students cover the costs of tuition.
* They have a rich history and tradition.
* They are often involved in research and innovation.
```

```
--- RAG (With your Knowledge Graph) ---
Processing SPARQL...

[Attempt 1] Executing:
Error: HTTPConnectionPool(host='localhost', port=11434): Read timed out. (read timeout=60)
Error: Expected {SelectQuery | ConstructQuery | DescribeQuery | AskQuery}, found 'Error'  (at char 0), (line:1, col:1)

[Attempt 2] Executing:
SELECT ?s ?p ?o WHERE {
 ?s ?p ?o . FILTER(CONTAINS(LCASE(STR(?s)), "university") || CONTAINS(LCASE(STR(?o)), "university")) }

Final Answer (Grounded):
Sure, here's the information you requested:

**Entity: Haigazian University**
* Property: P646 | Value: http://www.wikidata.org/entity/statement/Q465523-032BC442-6C56-425B-853E-D4B5AE319C58

**Entity: Bursa Uludağ University**
* Property: comment | Value: Local entity not found in Wikidata

**Entity: Lebanese American University**
* Property: P373 | Value: http://www.wikidata.org/entity/statement/Q464361-1772990D-7D5D-4B87-B668-A9BF09C4B274

**Entity: Case Western Reserve University**
* Property: P31 | Value: http://www.wikidata.org/entity/statement/q1047060-1FDE7921-248B-42BC-80FA-9AEC2AF25725

**Entity: Case Western Reserve University**
* Property: altLabel | Value: 凱斯西方儲備大學

**Entity: Haigazian University**
* Property: identifiers | Value: 21

**Entity: Cambridge University Press**
* Property: description | Value: brita eldonejo

**Entity: Lebanese American University**
* Property: P2003 | Value: http://www.wikidata.org/entity/statement/Q464361-1772990D-7D5D-4B87-B668-A9BF09C4B274

**Entity: - Saint Joseph University**
* Property: comment | Value: Local entity not found in Wikidata

**Entity: Harvard university**
```

```
**Entity: - Saint Joseph University**
* Property: comment | Value: Local entity not found in Wikidata

**Entity: Harvard university**
* Property: description | Value: 一所位於美國麻省劍橋市的大學

**Entity: Stanford University**
* Property: comment | Value: Local entity not found in Wikidata

**Entity: the American University of Beirut 18**
* Property: type | Value: http://www.w3.org/2000/01/rdf-schema#Resource

**Entity: University of Balamand**
* Property: P131 | Value: http://www.wikidata.org/entity/Q168954

**Entity: Case Western Reserve University**
* Property: description | Value: アメリカの私立大学

Your question (or 'exit'):
```

The RAG method succeeded to give us good answers. Indeed, Lebanese American University (LAU), Saint-Joseph University (USJ), American University of Beirut (AUB) and University of Balamand are true universities, located in Lebanon.

Third test: Tell me about American university

```
Your question (or 'exit'): tell me about american university

--- BASELINE (LLM only, no KG) ---
Sure, here's a summary of American universities:

**What are American universities?**

American universities are institutions of higher education that offer undergraduate and graduate programs in various fields of study. They are accredit
ed by regional and national agencies, such as the Southern Association of Colleges and Universities (SACU) and the North Central Association of Schools
 of Medicine (NCAOM), to ensure that students receive a quality education.

**Key characteristics of American universities:**

* **Research-oriented:** American universities are known for their strong research programs, with many departments and labs focusing on cutting-edge re
search in various fields.
* **Diverse student body:** American universities attract students from diverse backgrounds, including international students.
* **Academic freedom:** American universities offer a wide range of academic programs and courses, allowing students to pursue their interests and expl
ore their passions.
* **Student life:** American universities provide a vibrant student life with various clubs, organizations, and activities to choose from.
* **High graduation rates:** American universities have a high graduation rate, with graduates finding employment in various industries and fields.

**Some of the most famous American universities include:**

* Harvard University
* Yale University
* Stanford University
* Princeton University
* Massachusetts Institute of Technology (MIT)
* University of California, Berkeley
* University of California, Los Angeles

**Benefits of attending an American university:**

* Access to a wide range of academic programs and resources
* Opportunities for research and internships
* Exposure to a diverse and stimulating learning environment
* Potential for future career success

**Note:** The term "university" is often used interchangeably with "college," but in the United States, universities offer more advanced degrees such a
s master's and doctoral programs.
```

```
--- RAG (With your Knowledge Graph) ---
Processing SPARQL...

[Attempt 1] Executing:
?s ?p ?o .

FILTER (
  CONTAINS (
    STR(?s),
    "American University"
  )
  || CONTAINS (
    STR(?o),
    "American University"
  )
) .

?s ?p ?o .
Error: Expected {SelectQuery | ConstructQuery | DescribeQuery | AskQuery}, found '?'  (at char 0), (line:1, col:1)

[Attempt 2] Executing:
SELECT ?s ?p ?o WHERE { ?s ?p ?o . FILTER(CONTAINS(LCASE(STR(?s)), "american university") || CONTAINS(LCASE(STR(?o)), "american university")) }

Final Answer (Grounded):
**Information about American University:**

* **Entity:** American University of Beirut
* **Property:** label
* **Value:** The American University of Beirut
* **URL:** <https://wikidata.org/wiki/Q469482>

**Additional facts about American University:**

* It is a secular and private American university located in Beirut, Lebanon.
* It is one of the oldest and most prestigious universities in the Arab world.
* It has a rich history of academic excellence and research.
* It is home to a wide range of students and faculty from around the world.
* It is a major publisher of scholarly journals and books.

Your question (or 'exit'):
```

The model succeeded again and give us the right answer thanks to our RDF Graph, as he mentioned the American University of Beirut and its description.

Fourth test: Tell me about museums in Beirut

```
Your question (or 'exit'): tell me about museums in beirut

--- BASELINE (LLM only, no KG) ---
Sure, here's a summary of museums in Beirut:

**The National Museum of Beirut:**

* Located in the heart of the city, this museum houses a vast collection of artifacts from ancient times to the present.
* Highlights include Roman mosaics, Byzantine paintings, and a rich collection of Lebanese art.
* Offers guided tours and exhibitions, providing insights into Beirut's rich history and culture.

**The Museum of Islamic Art:**

* Located in the heart of the Gemmay district, this museum focuses on Islamic art and artifacts from the Middle East and North Africa.
* Features a stunning collection of mosaics, ceramics, and textiles from the 7th to the 13th centuries.
* Offers educational programs and workshops, promoting understanding and appreciation of Islamic art.

**The Museum of Lebanese History:**

* Located in the Achrafieh neighborhood, this museum showcases the history of Lebanon from prehistoric times to the present.
* Highlights include archaeological finds, traditional crafts, and a rich collection of paintings and sculptures.
* Offers guided tours and exhibitions, providing a glimpse into Lebanon's fascinating past.

**The Museum of Modern Art (MoMA):**

* The MoMA houses a diverse collection of modern and contemporary art from Lebanese and international artists.
* Features works by renowned artists such as Picasso, Matisse, and Warhol.
* Offers a unique and contemporary perspective on Lebanese art and culture.

**The Beirut History Museum:**

* Located in the heart of the city, this small museum offers a glimpse into Beirut's history from the 19th to the 20th centuries.
* Features exhibits on the city's political, social, and economic development.
* Offers a valuable resource for understanding Beirut's past.

**Additional Tips:**

* To ensure a more enriching experience, consider booking guided tours or attending cultural events and workshops organized by museums.
* Many museums offer free admission on certain days or hours.
* Public transportation is readily available, making it easy to get around the city.
```

```
--- RAG (With your Knowledge Graph) ---
Processing SPARQL...

[Attempt 1] Executing:
SELECT ?s ?p ?o WHERE {
  ?s ?p ?o .
  FILTER(CONTAINS(LCASE(STR(?s)), "museum"))
}

Final Answer (Grounded):
## Museums in Beirut

Here's the information about museums in Beirut from the RDF triples:

**1. The Museum of Fine Arts**

* Entity: `the Museum of Fine Arts`
* Property: `description`
* Value: `museum pertunjukan seni di Jepang`
* Wikidata link: `http://www.wikidata.org/entity/statement/Q5359197-576E6F0E-46D5-4EAE-A3A5-7115D1B8F281`

**2. Mim Museum**

* Entity: `Mim Museum`
* Property: `description`
* Value: `museum of minerals and fossils in Beirut, Lebanon`
* Wikidata link: `http://www.wikidata.org/entity/Q3820`

**3. The Museum of Fine Arts**

* Entity: `the Museum of Fine Arts`
* Property: `P17`
* Value: `http://www.wikidata.org/entity/statement/Q5359197-4AE04275-D16D-47BD-88DD-4B9A94003D7E`
* Wikidata link: `http://www.wikidata.org/entity/Q207694`

**4. The Museum of Fine Arts**

* Entity: `the Museum of Fine Arts`
* Property: `P18`
* Value: `http://www.wikidata.org/entity/statement/Q5359197-1582d55f-4405-14f9-d256-0eba50360184`
* Wikidata link: `http://www.wikidata.org/entity/Q5359197-1582d55f-4405-14f9-d256-0eba50360184`
* Wikidata link: `http://www.wikidata.org/entity/Q207694`

**5. The Museum of Fine Arts**

* Entity: `the Museum of Fine Arts`
* Property: `label`
* Value: ` متحف الفنون الجميلة (الايان)`
* Wikidata link: `http://www.wikidata.org/entity/Q3000005`

**6. The Museum of Fine Arts**

* Entity: `the Museum of Fine Arts`
* Property: `description`
* Value: `kunstmuseum in Gifu, Japan`
```

Once again, the model succeeded to give us answers thanks to our RDF Graph.

Fifth test: give me information about Lebanon

```
--- BASELINE (LLM only, no KG) ---
Sure, here's some information about Lebanon:

**Geography:**

* Lebanon is located in the Western Asia region.
* It is bordered by Israel to the north, Jordan and Syria to the east, and the Mediterranean Sea to the west.
* The country has a diverse landscape, with mountains, valleys, and coastal plains.

**History:**

* Lebanon has a rich and complex history.
* It was once a major power in the Eastern Mediterranean, and played a significant role in the development of the Arab world.
* Lebanon fought in World War I and World War II, and was a founding member of the United Nations.
* The country gained independence in 1945, but faced numerous challenges in the following decades, including war, political instability, and economic d
ifficulties.

**Culture:**

* Lebanon is a vibrant and diverse country with a rich cultural heritage.
* The country is known for its traditional music, dance, and cuisine.
* Lebanon is also a major center for Christian tourism, with many historical sites and churches to explore.

**Government:**

* Lebanon is a constitutional republic with a president as head of state and a prime minister as head of government.
* The current president is Michel Aoun.

**Economy:**

* Lebanon is a heavily dependent on tourism, trade, and investment.
* The country is also a major producer of olive oil, wine, and citrus fruits.
* Lebanon faces significant challenges in terms of unemployment, poverty, and corruption.

**Languages:**

* Lebanon is a bilingual country, with Arabic and French being the official languages.
* Other languages spoken in Lebanon include English, Greek, and Syrian.

**Other facts:**

* Lebanon is a popular destination for hiking, skiing, and diving.
* The country is home to a variety of wildlife, including birds, fish, and reptiles.
* Lebanon is a major cultural and artistic hub in the Middle East.
```

```
--- RAG (With your Knowledge Graph) ---
Processing SPARQL...

[Attempt 1] Executing:
?s ?p ?o .

# Lebanon
?s ?p/?p/?p ?o .

# Information about Lebanon
?s ?p/?p/?p ?o .
```
Error: Expected {SelectQuery | ConstructQuery | DescribeQuery | AskQuery}, found '?' (at char 0), (line:1, col:1)

```
[Attempt 2] Executing:
SELECT ?s ?p ?o WHERE {
 ?s ?p ?o . FILTER(CONTAINS(LCASE(STR(?s)), "lebanon") || CONTAINS(LCASE(STR(?o)), "lebanon")) }

Final Answer (Grounded):
**Information about Lebanon:**

* **Entity:** The Bank of Lebanon (http://www.w3.org/2000/01/rdf-schema#Resource)
* **Property:** type (http://www.w3.org/2000/01/rdf-schema#Resource) - Resource
* **Value:** http://www.w3.org/2000/01/rdf-schema#Resource
* **Entity:** Byblos (Mediterranean city in the Mount Lebanon Governorate, Lebanon, a main Canaanite-Phoenician city) (http://www.wikidata.org/entity/Q
2915324)
* **Property:** description (http://www.w3.org/2000/01/rdf-schema#Description) - Mediterranean city in the Mount Lebanon Governorate, Lebanon, a main C
anaanite-Phoenician city
* **Entity:** Mount Lebanon (Núi Liban) (http://www.wikidata.org/entity/Q130842)
* **Property:** label (http://www.w3.org/2000/01/rdf-schema#Label) - Núi Liban
* **Entity:** Q163490 (kota pesisir yang terletak di Lebanon bagian selatan) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Q2915324 (pinggir kota di Lebanon) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Q130842 (French mandate of Lebanon) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Mount Lebanon (Libanonfjellene) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Mount Lebanon (Mohafazat Mont-Liban) (http://data.europa.eu/resource/id/123456789)
* **Entity:** MTV Lebanon (Prime Minister of Lebanon, businessman) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Greater Lebanon (Greater Lebanon) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Mount Lebanon (레바논산맥) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Q6538216 (onye na-eme ihe nkiri si Lebanon) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Greater Lebanon (Greater Lebanon) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Mount Lebanon (Leibalona) (http://data.europa.eu/resource/id/123456789)
* **Entity:** Q192778 (Prime Minister of Lebanon, businessman) (http://data.europa.eu/resource/id/123456789)
```

The model succeeded again in giving good answers thanks to our database. Indeed, he mentioned a lot of places related to Lebanon such as banks (Bank of Lebanon), Cities (Byblos), TV Channels ect.

VIII) Conclusion

In this project we learned how to scrap data and collect data. Moreover, we explored the complete process of building an RDF Graph. Indeed, we collected the data, aligned it with WIKIDATA and finally we constructed a chatbot that use the RAG in order to query our database for answering the questions.

Moreover, we also understood that the results qualities of such system highly depend on the data quality, model configuration and training conditions. Here, in our study, the model TransE appeared to be the better choice.

Finally, we succeeded to create a functional chatbot that gives answers to questions about Lebanon. Now, you know everything about Lebanon, you just have to ask the chatbot based on ollama model and our RDF Graph!